Technical Feasibility and Architectural Assessment for Space-Based Alpine Flood and Debris Mapping

Physical Principles and Sensor Trade-Offs in Mountainous Terrain

Synthetic Aperture Radar versus Optical Spectral Signatures

Spaceborne emergency response in steep alpine environments, such as the Bhote Koshi-Trishuli corridor in northern Nepal, requires an understanding of how electromagnetic waves interact with mountain surface cover[1]. Active C-band Synthetic Aperture Radar (SAR), such as the sensor onboard Sentinel-1 operating at a central frequency of $5.405\ \text{GHz}$ ($\lambda \approx 5.55\ \text{cm}$), emits microwave signals that penetrate atmospheric cloud cover, monsoonal rain, and smoke[1]. Sentinel-1 measures both backscatter amplitude (σ^0) and phase across dual-polarization channels (VV and VH) at a $10\ \text{m}$ spatial resolution in Interferometric Wide (IW) Ground Range Detected (GRD) mode[3].

Clear standing water acts as a smooth dielectric interface relative to C-band wavelengths, inducing specular reflection that directs incident microwave pulses away from the radar sensor[3]. This yields characteristically low backscatter intensity values, typically falling below $-20\ \text{dB}$ [3]. Conversely, mass-movement deposits—including saturated debris, rock avalanches, mudslides, and scoured gravel channels—exhibit high surface roughness and dielectric variation[3]. These rough surfaces trigger strong diffuse or volume scattering, raising backscatter intensities to between $-10\ \text{dB}$ and $0\ \text{dB}$ [3].

Sentinel-2 MultiSpectral Instrument (MSI) optical sensors capture solar radiation reflected from the Earth's surface across 13 spectral bands spanning the visible, near-infrared (NIR), and shortwave infrared (SWIR) wavelengths[4]. Optical mapping relies on surface material absorption and reflectance properties rather than geometry or dielectric constants[3]. The Normalized Difference Water Index ($\text{NDWI} = (\rho_{\text{Green}} - \rho_{\text{NIR}})/(\rho_{\text{Green}} + \rho_{\text{NIR}})$) and Modified Normalized Difference Water Index ($\text{MNDWI} = (\rho_{\text{Green}} - \rho_{\text{SWIR}})/(\rho_{\text{Green}} + \rho_{\text{SWIR}})$) isolate open water by exploiting high green reflectance and strong NIR/SWIR absorption[3].

Secondary water-quality indices further distinguish hyper-concentrated sediment flows from clear water[3]. The Normalized Difference Turbidity Index (

$\mathrm{NDTI} = (\rho_{\mathrm{Red}} - \rho_{\mathrm{Green}})/(\rho_{\mathrm{Red}} + \rho_{\mathrm{Green}})$) and the Normalized Difference Chlorophyll Index ($\mathrm{NDCI} = (\rho_{\mathrm{RedEdge1}} - \rho_{\mathrm{Red}})/(\rho_{\mathrm{RedEdge1}} + \rho_{\mathrm{Red}})$) measure suspended solids and organic material[3]. While optical spectral signatures can differentiate turbid water, exposed mud, and damaged canopy cover with fine spectral resolution, optical sensors are frequently blinded by monsoonal cloud cover during active disaster events[1].

Sensor Parameter / Capability	Sentinel-1 SAR (C-Band)	Sentinel-2 MSI (Optical)
Primary Physical Interaction	Surface roughness, structural geometry, and dielectric properties[3]	Solar reflectance and material spectral absorption[3]
Atmospheric & Night Capability	Full cloud, haze, and precipitation penetration; day/night operational[1]	Severe cloud and fog degradation; daytime operational only[1]
Spatial Resolution	$10\ \mathrm{m} \times$ (IW GRD spatial sampling)[4]	$10\ \mathrm{m}$ (Visible/NIR), $20\ \mathrm{m}$ (Red Edge/SWIR), $60\ \mathrm{m}$ (Atmospheric)[4]
Inundated Clear Water Return	Low backscatter ($\sigma^0 <$) via specular reflection[3]	High absorption in NIR/SWIR; strongly positive NDWI/MNDWI[3]
Debris Flow / Mud Signature	Moderate-to-high backscatter ($\sigma^0 >$) via rough diffuse scattering[3]	High Red/SWIR reflectance; elevated NDTI (turbidity)[3]
Primary Physical Artifacts	Radar shadow, layover, foreshortening, specular wet-mud returns[3]	Cloud obscuration, topographic shadows, atmospheric haze[7]

Geometric and Radiometric Normalization Requirements

Pixel-based change detection across pre-disaster and post-disaster satellite acquisitions requires geometric co-registration and radiometric standardization[1]. Sentinel-1 operates with a side-looking radar geometry at fixed off-nadir look angles ranging from 30° to 45° [5].

Comparing acquisitions from different relative orbit tracks (e.g., an ascending track looking east-northeast versus a descending track looking west-southwest) introduces geometric parallax[1]. Steep mountain ridges tilt in opposite directions relative to the satellite's line of sight, creating severe relief displacements ($\Delta r = h \cdot \cot \theta_i$) that render two-dimensional spatial registration mathematically impossible[5]. Automated change detection must therefore restrict image comparisons strictly to identical Relative Orbit Tracks acquired under matching geometry[1].

Radiometric comparability requires Radiometric Terrain Flattening (RTC)[1]. Standard radar backscatter metrics (σ^0) assume a flat ground surface, causing steep terrain slopes facing the radar sensor to appear artificially bright due to a larger local contributing ground area per range cell[3]. Converting Ground Range Detected (GRD) intensity values to gamma-nought radiometric terrain-flattened backscatter (γ^0_{RTC}) corrects for local illuminated area variations using a high-resolution Digital Elevation Model (such as the Copernicus WorldDEM-30)[1]. Thermal noise removal must also be applied to eliminate instrument noise in cross-polarization (VH) channels at low signal levels, ensuring that temporal changes reflect physical ground alterations rather than orbital or processing artifacts[3].

Topographic and Environmental Artifacts in Alpine Corridors

High-relief alpine environments introduce physical distortion mechanisms that systematically degrade conventional machine learning classifiers:

- **Radar Shadow and Layover**: When mountain slopes face away from the radar look direction at angles steeper than the sensor's look angle, no microwave energy reaches the ground, creating radar shadow[3]. Shadowed pixels record only instrument noise, displaying low backscatter values ($\sigma^0 < -22\ \mathrm{dB}$) that match the specular reflection of open water[3]. This produces widespread false-positive flood detections along steep north-facing slopes. Conversely, layover occurs on steep slopes facing the sensor, where the radar wavefront strikes the mountain peak before the base, mapping mountain tops onto valley floors and corrupting spatial boundaries[5].
- **Vegetation Canopy Scattering**: Dense forest canopies induce volume scattering[3]. Monsoonal rainfall increases canopy water content and changes dielectric constants, elevating backscatter returns by $2\ \mathrm{dB}$ to $5\ \mathrm{dB}$. This canopy noise can obscure sub-canopy inundation or mimic ground-level debris accumulation[3].
- **Specular Reflection from Saturated Mud**: High-velocity debris flows leave smooth deposits of fine, saturated silt along valley floors. These wet, fine-grained mud surfaces undergo specular reflection, generating low backscatter signatures identical to standing water[3]. Consequently, single-date SAR backscatter intensity alone cannot differentiate standing water from smooth mud deposits without integrating elevation priors or temporal optical data[3].
- **Speckle Noise**: Coherent wave interference produces multiplicative speckle noise,

causing granular intensity fluctuations across homogeneous targets[9]. Speckle corrupts pixel-level neural feature extractors unless suppressed using spatio-temporal filtering (such as Refined Lee or multitemporal stack filtering)[9].

- **Mixed Pixel Contamination**: Narrow Himalayan river channels ($10\ \mathrm{m}$ to $30\ \mathrm{m}$ wide) frequently span multiple land-cover classes within a single $10\ \mathrm{m}$ pixel grid[4]. These mixed pixels combine turbid water, exposed bedrock, destroyed vegetation, and road remnants, blurring class boundaries and causing edge classification dropouts[4].
- **Acquisition Timing Latency**: Sentinel-1 operates on a 12-day single-satellite repeat cycle (or 6 days when a constellation pair is active)[1]. Because high-altitude glacier collapses, landslide dam bursts, and flash floods evolve over minutes to hours, satellite passes rarely coincide with peak discharge[1]. Imagery captured days after an event often records receding water levels, dried mud crusts, or re-established channels rather than peak inundation, leading to underestimation of maximum hazard extents[3].

Task Formulation and Training Data Compatibility Audit

Disambiguating Inundation, Debris Flow, and Infrastructure Damage

Accurate post-disaster mapping requires distinguishing between three distinct physical hazards, each characterized by unique scattering mechanisms and spectral profiles:

Inundation mapping delineates open, standing, or slow-moving water[1]. Its primary physical indicator is a drop in SAR backscatter intensity ($\Delta\sigma^0 < -4\ \mathrm{dB}$) caused by specular reflection, paired with high optical NDWI and MNDWI values[3].

Debris and sediment detection identifies rockslides, ice-mud avalanches, and sediment deposits along valley floors[5]. In SAR imagery, coarse rock deposition increases micro-relief roughness, causing backscatter increases ($\Delta\sigma^0 > +3\ \mathrm{dB}$ to $+8\ \mathrm{dB}$) relative to pre-event baselines[3]. In optical imagery, debris flows strip away vegetation, replacing high near-infrared returns with high red/SWIR reflectance and elevated NDTI turbidity values[3].

Infrastructure damage classification detects structural destruction across roads, bridges, and settlements[1]. Built structures act as corner reflectors that generate strong double-bounce backscatter in pre-event SAR imagery[5]. Severe structural damage eliminates double-bounce geometry, causing localized backscatter drops or InSAR coherence decorrelation[5]. Because individual buildings and road segments are narrow relative to $10\ \mathrm{m}$ satellite pixels, direct pixel-level semantic segmentation of infrastructure damage is prone to high error rates[4]. Reliable damage assessment requires intersecting high-confidence hazard spatial masks (flood and debris) with pre-existing vector infrastructure networks[1].

Audit of KuroSiwo and Sen1Floods11 Datasets

Applying public deep learning benchmarks to high-relief Himalayan disaster events introduces

domain shift that can compromise model performance[1].

Evaluation Parameter	KuroSiwo Dataset	Sen1Floods11 Dataset	Target Himalayan Domain
Primary Focus	Multi-temporal SAR flood mapping[9]	Global Sentinel-1 flood mapping[4]	Compound flood, debris, and settlement isolation[1]
Topographic Representation	Lowland plains, coastal deltas, river basins[9]	Flat agricultural plains, lowland river basins[4]	Extreme alpine relief, steep V-shaped valleys[1]
Target Physical Process	Open-water inundation (σ^0 drops)[3]	Open-water inundation (σ^0 drops)[3]	High-velocity debris flows, mud deposits, flooding[5]
Label Noise & Verification	Low noise; expert manual annotations[11]	High noise in weak split; auto JRC thresholded[4]	Mixed signal signatures, topographic shadows[3]
Himalayan Generalization	Poor for debris; susceptible to shadow noise[3]	Poor; weak labels corrupt terrain features[3]	Requires topographic normalization and domain adaptation[1]

The KuroSiwo dataset contains global multi-temporal Sentinel-1 SAR imagery (GRD and Single Look Complex patches) with expert manual annotations[9]. Although KuroSiwo provides high label accuracy, its geographical sampling heavily favors lowland floodplains, coastal wetlands, and broad river valleys[9]. It models water almost exclusively as a specular reflection surface (σ^0 drops)[3]. It lacks explicit classes for high-velocity alpine rock-ice avalanches, sediment-laden mudflows, or mountain channel scouring where radar backscatter increases[3]. Models trained solely on KuroSiwo fail to detect high-roughness debris deposits, misclassifying them as unaffected land.

The Sen1Floods11 dataset includes Sentinel-1 and Sentinel-2 chips across 11 global flood events[4]. Its weakly supervised training split was generated automatically by thresholding historical Joint Research Centre (JRC) Permanent Water layers and Sentinel-1 backscatter[4]. In steep mountain terrain, radar shadows and cloud cover artifacts contaminate these weakly supervised labels with false-positive water annotations[3]. Furthermore, its hand-labeled test set contains very few alpine scenes, making it insufficient for training models on mountain hazard

cascades[4].
Public training benchmarks model flooding as a planar water-extent task over gentle terrain[4]. Himalayan events, such as the Trishuli or Chamoli disasters, represent dynamic hazard cascades involving rock avalanches, mud deposition, and channel blockages[5]. Deploying models trained on KuroSiwo or Sen1Floods11 without domain adaptation results in false negatives along debris tracks and false positives in radar shadow zones[3].

Zero-Leakage Evaluation Protocol for Alpine Corridors

Evaluating model generalization to unseen Himalayan scenes requires strict isolation between training, validation, and test splits[1].
Random patch-based or tile-based splits across the same satellite image introduce severe spatial autocorrelation[1]. Adjacent 512×512 pixel patches share structural terrain features, atmospheric conditions, and speckle noise, leading to artificially inflated evaluation scores. To prevent spatial leakage, data splitting must enforce a Geographic Catchment Isolation Protocol[1]. The training split must be assembled from geographically distinct river basins (such as the Marsyangdi, Seti River, or historical 2021 Chamoli catchments)[5]. The evaluation set must consist of completely disjoint river corridors (such as the Trishuli-Bhote Koshi corridor) separated by a spatial buffer exceeding $50\ \mathrm{km}$ [1].
Models must not access pre-event imagery or orbit tracks from the evaluation catchment during training[1]. Relative Orbit Tracks assigned to the validation set must be completely excluded from training splits to prevent neural networks from memorizing static, terrain-induced backscatter patterns or specific shadow geometries[5].
Dataset normalization statistics (mean, standard deviation, feature scaling) must be computed strictly on the training split[1]. Elevation-derived features (such as slope, aspect, and HAND) must be generated independently per catchment to prevent global spatial statistics from leaking across splits[1].
Spatial Block Cross-Validation must partition training and validation folds across major Himalayan river basins:

- **Fold 1**: Training on the Western Himalaya (Annapurna/Gandaki basins); Validation on the Central Himalaya (Chamoli/Alaknanda basins)[5].
- **Fold 2**: Training on the Central Himalaya (Karnali/Seti basins); Validation on the Marsyangdi basin.
- **Fold 3**: Training on the Eastern Himalaya (Tamur/Arun basins); Validation on the Trishuli-Bhote Koshi corridor[1].

Multimodal Data Fusion and Network Graph Analytics

Hydro-DEM Topographic Filtering

Raw semantic segmentation predictions from satellite classifiers often contain false positives, such as predicted water bodies on steep mountain ridges or isolated flood pixels in high-altitude canopy cover. Incorporating elevation data provides physical constraints that eliminate these

non-physical predictions[1].

Using the Copernicus WorldDEM-30 ($30\ \mathrm{m}$ spatial resolution), three hydrologically relevant spatial rasters are computed[1]:

- **Slope Map (θ_{slope})**: Local topographic gradient derived via spatial elevation derivatives:

$$\theta_{\mathrm{slope}} = \arctan\left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\right)$$

- **Height Above Nearest Drainage (HAND)**: Normalizes terrain elevation relative to the local drainage network along hydraulic flow paths[10]. For every pixel, HAND measures the vertical distance Δz to the nearest stream channel cell determined by flow accumulation lines.
- **Flow Accumulation**: Calculates the upstream contributing area per cell using deterministic 8-direction (D8) flow routing algorithms.

Physically constrained rules refine the classifier predictions:

- **Inundation Spatial Rule**: Standing water cannot persist on steep slopes or elevated terrain[10]. Pixels classified as flooded are masked out as false positives if $\theta_{\mathrm{slope}} >$ or if $\mathrm{HAND} > 15\ \mathrm{m}$, unless confirmed as an established glacial lake.
- **Debris Flow Spatial Rule**: Debris deposits concentrate along valley floors and channel corridors[5]. Candidate debris predictions are validated only if they lie within low HAND zones ($\mathrm{HAND} < 30\ \mathrm{m}$) or along high flow-accumulation channels downslope from identified slope failures[5].

Vector Network Graph Topology and Isolation Analysis

To identify cut-off settlements, the system builds a road network graph $G = (V, E)$ from OpenStreetMap (OSM) snapshots acquired prior to the event date[1].

Graph construction proceeds as follows:

- **Graph Nodes (V)**: Represent road intersections, bridge structures, village settlements, district towns, and regional medical facilities[1].
- **Graph Edges (E)**: Represent road segments connecting nodes, weighted by physical length L_e and nominal travel time T_e [13].

AI-derived flood and debris spatial masks are converted into spatial vector polygons P_{hazard}[1]. Spatial intersection queries identify damaged infrastructure segments:

$E_{\text{damaged}} = \{e \in E \mid \text{ST_Intersects}(e, P_{\text{hazard}})\}$. Edge traversal weights are updated dynamically:

$$W(e) = \begin{cases} \infty & \text{if } e \in E_{\text{damaged}} \text{ (Road/Bridge Severed)} \\ T_e & \text{if } e \notin E_{\text{damaged}} \end{cases}$$

Let $S \subset V$ represent populated village nodes, and $H \subset V$ represent functional emergency facilities (such as district hospitals or relief centers)[1]. For every settlement $s \in S$, pathfinding algorithms (such as Dijkstra or A*) compute post-disaster reachability to the nearest accessible medical facility $h^* \in H$:

$$d_{G'}(s, H) = \min_{h \in H} \text{ShortestPath}(G', s, h)$$

A settlement s is flagged as **Cut-Off** if $d_{G'}(s, H) = \infty$, indicating complete network disconnectivity[1].

System Architecture and Operational Pipeline

End-to-End Operational Pipeline Specifications

The automated processing system ingests raw satellite imagery and elevation models, executing multi-stage feature extraction, topological filtering, graph analytics, uncertainty quantification, and report generation[1].

Pipeline Module	Input Data	Processing Algorithm / Core Framework	Output Data / Artefact
1. Data Ingestion & Preprocessing	Raw Sentinel-1 GRD, Sentinel-2 L2A, Copernicus DEM[1]	SNAP/GDAL processing: Doppler geocoding, Same-Orbit Alignment, γ^0_{RTC} flattening, Sen2Cor cloud masking[1]	Co-registered, radiometrically calibrated image stacks ($10\ \text{m}$ grid)[1]

2. Multimodal Neural Segmentation	Pre/Post S1 & S2 image stacks, DEM slope/aspect[1]	Dual-Stream ResNet/Swin Transformer backbone with Temporal Difference Attention Cross-Fusion[11]	Softmax class probability maps: $P(\text{Water})$, $P(\text{Debris})$, $P(\text{Background}$
3. Hydro-DEM Filtering	Softmax maps, Copernicus DEM (30m)[1]	D8 hydrologic routing, HAND raster generation, slope thresholding ($\theta_{\text{slope}} >$)[10]	Refined binary and multi-class vector hazard masks (P_{hazard})[1]
4. Topological Graph Engine	P_{hazard}, Pre-event OSM Snapshot (ohsome API)[1]	NetworkX/PostGIS spatial intersection, edge deletion, Dijkstra reachability pathfinding[13]	Damaged road/bridge segments, list of isolated village nodes S_{isolated} [cite: 1, 2]
5. Conformal Uncertainty Quantification	Softmax probability maps, Calibration Set D_{cal} [cite: 15, 16]	Class-Conditional Conformalized Risk Control (CRC)[16]	Spatial uncertainty heatmaps, coverage-guaranteed prediction sets $C_\alpha(X)$ [cite: 15, 16]
6. Grounded Copilot Engine	S_{isolated}, damaged infrastructure vectors, UQ metrics[1]	Deterministic JSON context aggregation + Large Language Model constrained by strict schema[1]	Structured JSON summary, bilingual situation report (English & Nepali)[1]

Situation-Report Copilot and Grounded Generation

The Situation-Report Copilot converts spatial graph metrics into concise situation reports for emergency managers[1]. To prevent hallucinated numbers or unverified claims, text generation is strictly decoupled from spatial metric calculations[1]:

Spatial analysis engines compute all quantitative metrics (such as flooded surface area in km^2, length of damaged roads in km, count of damaged bridges, and exact names of isolated

villages) using deterministic spatial queries[1]. These values are assembled into a structured JSON payload.

The Large Language Model receives this JSON payload alongside system prompts that strictly forbid generating any number, location name, or casualty figure not present in the input context[1]. System prompts enforce bilingual text generation in both English and Nepali[1]. Nepali language output uses standardized disaster management terminology approved by the Nepal National Disaster Risk Reduction and Management Authority (NDRRMA)[1].

Uncertainty Quantification and Anti-Hallucination Controls

Conformal Prediction for Risk-Controlled Semantic Segmentation

To prevent unverified hazard claims from misleading rescue teams, the neural network must output statistical uncertainty bounds alongside classification predictions[15]. Standard softmax outputs are often overconfident and do not provide statistical guarantees[15]. The system implements Split Conformal Prediction (CP) and Conformalized Risk Control (CRC) to generate prediction sets with user-defined coverage guarantees[15].

Let $X \in \mathbb{R}^{C \times H \times W}$ represent the input satellite image stack, and $Y \in \{1, \dots, K\}^{H \times W}$ represent the true ground-truth segmentation mask across K classes (where $1 = \text{Background}$, $2 = \text{Water}$, and $3 = \text{Debris}$)[16]. The neural network outputs estimated class probability vectors $\hat{f}(X)_{ij} \in [0,1]^K$ for each pixel position (i, j) [16].

A calibration dataset $D_{\text{cal}} = \{(X_n, Y_n)\}_{n=1}^{N}$, independent of training and testing sets, is evaluated[16]. For each pixel, a non-conformity score $S_{n,ij}$ measures prediction error[16]:

$$S_{n,ij} = 1 - \hat{f}(X_n)_{ij,Y_{n,ij}}$$

To control false negatives in disaster detection, Class-Conditional Conformal Prediction (CCCP) derives separate quantile thresholds for each class $k \in \{1, \dots, K\}$ [17]. Given a user-defined risk tolerance $\alpha \in (0,1)$ (such as $\alpha = 0.05$, corresponding to a 95% statistical coverage guarantee)[15], the class conformal threshold $\hat{q}_{\alpha,k}$ is calculated as the empirical $\frac{\lceil (N_k+1)(1-\alpha) \rceil}{N_k}$ quantile of the non-conformity scores computed over calibration

pixels belonging to class k [17].

For a new test pixel x at position (i, j), the system constructs a Prediction Set $C_\alpha(x) \subseteq \{1, \ldots, K\}$ [16]:

$$C_\alpha(x) = \left\{ k \in \{1, \ldots, K\} \mid \hat{f}(x)_{ij,k} \geq 1 - \hat{q}_{\alpha,k} \right\}$$

Under data exchangeability, the prediction set $C_\alpha(x)$ satisfies the marginal coverage guarantee[16]:

$$P\left(Y_{\text{test}} \in C_\alpha(X_{\text{test}})\right) \geq 1 - \alpha$$

Practical Interpretation of Prediction Sets

The size (cardinality) of the prediction set $|C_\alpha(x)|$ directly reflects predictive uncertainty[16]:

- **Singular Prediction Set ($|C_\alpha(x)| = 1$)**: High model confidence. If $C_\alpha(x) = \{\text{Water}\}$, the pixel is confidently classified as open water[16].
- **Multi-Label Prediction Set ($|C_\alpha(x)| \geq 2$)**: High prediction ambiguity. If $C_\alpha(x) = \{\text{Water}, \text{Debris}\}$, the physical signature at that location is ambiguous (such as wet mud or shallow turbulent flow)[16].
- **Empty Prediction Set ($|C_\alpha(x)| = 0$)**: Out-of-distribution input anomaly (such as corrupt pixel values or extreme noise).

In the response dashboard, multi-label prediction sets are displayed as spatial uncertainty heatmaps. Regions of high ambiguity are highlighted, signaling to operational personnel that satellite-derived predictions in these zones require manual confirmation prior to dispatching emergency resources[16].

Evaluation Methodology Against Official Reference Maps

Evaluating system performance against official Copernicus Emergency Management Service (CEMS) reference maps (activation EMSR927 for the Trishuli flood event) requires a reproducible, zero-leakage benchmarking protocol[1].

Official CEMS vector polygons, UNOSAT damage maps, and post-event OSM edits are strictly

excluded from model input, feature engineering, and inference pipelines[1]. These layers are ingested solely as ground-truth evaluation targets[1].
CEMS EMSR927 vector layers delineating flood extents, mudslides, and damaged structures are rasterized onto a $10\ \mathrm{m}$ spatial grid matching the model's output tensor, aligned to the EPSG:32645 (UTM Zone 45N) coordinate reference system.

Evaluation Metric	Mathematical Formulation	Target Domain Focus
Class Intersection over Union (IoU_k)	$\mathrm{IoU}_k =$	Evaluated independently for Flood and Debris classes[14]
Boundary F1-Score ($\mathrm{F1}_{\mathrm{edge}}$)	$\mathrm{F1}_{\mathrm{edge}} =$	Evaluated within a $30\ \mathrm{m}$ buffer along vector hazard perimeters
Isolated Village Precision / Recall	$\mathrm{Precision}_{\mathrm{iso}} =$	Compares predicted cut-off nodes against official CEMS maps[1]

Architectural Alternatives, Failure Modes, and Falsification Framework

Comparative Analysis of Architectural Alternatives

Modeling Paradigm	Core Algorithmic Mechanism	Key Technical Strengths	Critical Vulnerabilities
Proposed: Dual-Stream Multimodal Transformer	Fusion of S1 SAR, S2 Optical, and DEM rasters using Swin Transformer backbones[1]	High accuracy; distinguishes open water from debris flows[9]	Requires GPU infrastructure; vulnerable to domain shift if uncalibrated[4]
Alternative A: Classical HAND-Otsu Thresholding	Automatic backscatter thresholding (Otsu) masked by HAND rasters[10]	Computationally efficient; deterministic; requires no deep learning[7]	High false-negative rate on rough debris; fails in radar shadow[3]

Alternative B: InSAR Coherence Change Detection	Interferometric phase coherence correlation from complex S1 SLC pairs[5]	Detects ground disruption at sub-pixel scales[5]	Severe temporal decorrelation over vegetation and snow[5]
Alternative C: Foundation Model Fine-Tuning	Pre-trained Satellite Image Time Series models (e.g., TiMo, Prithvi-EO-2.0)[14]	High data efficiency; robust zero-shot feature representations[9]	High computational footprint; spatial sampling fixed at 10m

Critical Failure Modes and Limitations

- **False Positives from Topographic Shadowing**: In steep, north-facing slopes, radar shadows attenuate backscatter down to the noise floor, triggering false positive water detections[3]. *Mitigation*: Mask all pixels where local incident angles satisfy $\theta_i < 5^\circ$ or where terrain shadow geometries projected from the Copernicus DEM intersect the line-of-sight vector[1].
- **False Negatives from Smooth Wet Mud Deposits**: Saturated silt deposits along riverbeds induce specular reflection, causing models to misclassify debris tracks as standing water[3]. *Mitigation*: Incorporate optical SWIR ratios and NDTI indices when clear-sky Sentinel-2 data is available; apply HAND slope rules ($\text{HAND} > 5\text{ m}$) to disallow standing water on sloped channels[3].
- **Sub-Pixel Channel Scouring and Resolution Loss**: Torrent channels narrower than 10 m blend with surrounding bank vegetation within a single 100 m^2 pixel cell, leading to omission of narrow flood paths[4]. *Mitigation*: Apply spatial super-resolution processing along OSM stream vectors prior to classification.
- **Graph Routing Errors from Incomplete Vector Maps**: Remote Himalayan valleys often contain unmapped trails or missing bridges in pre-event OpenStreetMap snapshots[1]. If a critical connecting segment is missing from the input graph, pathfinding algorithms will incorrectly flag reachable villages as isolated[1]. *Mitigation*: Ingest settlement density layers (such as the High-Resolution Settlement Layer) and run road-extraction overlays on pre-event optical imagery to repair network gaps prior to graph analysis.

Experimental Framework for Assumption Falsification

Four empirical experiments test and validate the core assumptions of this architecture:

Experiment 1: Testing Cross-Track Change Detection Performance

- **Hypothesis**: Pixel-based change detection across mixed ascending and descending orbit tracks produces elevated false-positive rates due to terrain relief displacement[1].
- **Setup**: Run the change detection pipeline on the Trishuli corridor using an ascending pre-event Sentinel-1 image (Track 56) paired with a descending post-event Sentinel-1

image (Track 63)[5]. Compare results against a control setup using matching orbit tracks (Track 56 pre- and post-event)[1].

- **Falsification Metric**: If the cross-track False Positive Rate (FPR) exceeds the same-track FPR by more than 15% over non-flooded steep terrain ($\theta_{\text{slope}} > 20^\circ$), the assumption that cross-track images can be used for change detection is empirically falsified.

Experiment 2: Evaluating All-Weather SAR Autonomous Mapping

- **Hypothesis**: Sentinel-1 SAR imagery alone, without optical Sentinel-2 confirmation, is sufficient to map complex alpine debris flows[1].
- **Setup**: Mask out post-event Sentinel-2 optical bands, forcing the network to segment debris flows using only dual-polarized Sentinel-1 SAR intensity and DEM rasters[1]. Evaluate mIoU against the complete multimodal pipeline (SAR + Optical + DEM) across the EMSR927 Trishuli validation area[1].
- **Falsification Metric**: If the SAR-only Debris Class IoU drops by more than 25% relative to the multimodal pipeline, the assumption of SAR self-sufficiency for debris mapping is falsified, proving optical data is necessary for resolving mud and rock deposits.

Experiment 3: Evaluating Weakly Supervised Pre-Training Benefits

- **Hypothesis**: Pre-training semantic segmentation backbones on global weakly supervised datasets (such as Sen1Floods11) improves generalization to Himalayan flood events compared to training from scratch[4].
- **Setup**: Train two identical network architectures: Model A initialized with weights pre-trained on Sen1Floods11 weakly supervised chips, and Model B initialized with ImageNet weights[4]. Fine-tune both models on a small set of manually annotated Himalayan scenes (e.g., Chamoli 2021) and evaluate on the unseen Trishuli 2026 dataset[1].
- **Falsification Metric**: If Model A fails to achieve a statistically significant improvement ($p < 0.05$) in mIoU over Model B, the utility of lowland weakly supervised pre-training for alpine flood mapping is falsified.

Experiment 4: Assessing Conformal Guarantees Under Domain Shift

- **Hypothesis**: Conformal prediction prediction sets maintain their 95% statistical coverage guarantee ($\alpha = 0.05$) when deployed across distant Himalayan catchments without local recalibration[15].
- **Setup**: Calibrate conformal quantile thresholds $\hat{q}_{\alpha,k}$ using a calibration set D_{cal} drawn from European flood scenes (KuroSiwo)[9]. Apply these fixed thresholds to generate prediction sets $C_\alpha(X)$ over the unseen Trishuli test scene[1]. Compute the true empirical

coverage rate $\text{Cov} = \frac{1}{|N_{\text{test}}|} \sum_{ij} \mathbb{I}(Y_{ij} \in C_\alpha(X_{ij}))$ [16].

- **Falsification Metric**: If the empirical coverage rate over the Trishuli test set falls below $1 - \alpha - 0.02$ (below 93% coverage for $\alpha = 0.05$), the assumption of domain-invariant conformal validity is falsified, proving that local alpine calibration sets are mandatory[16].

Nuanced Operational Recommendations

Operational deployment of space-based disaster mapping models in alpine terrain requires balancing satellite physical limits with user safety requirements[1].

First, because Sentinel satellites operate on multi-day repeat orbits, spaceborne systems cannot provide real-time early warnings for sudden glacier collapses minutes before impact[1]. Satellite mapping serves primarily as a post-event rapid damage assessment and routing tool[1].

Second, automated change detection pipelines must strictly enforce identical Relative Orbit Tracks and apply Radiometric Terrain Flattening (γ^0_{RTC}) to prevent terrain artifacts from corrupting hazard maps[1].

Third, language model copilots must be strictly constrained by deterministic spatial data payloads, preventing unverified numbers or casualty claims from entering emergency situation reports[1].

Finally, integrating Conformal Prediction heatmaps ensures that first responders receive transparent statistical confidence bounds alongside hazard extent maps, enabling safer and more informed emergency routing in complex mountain terrain[15].

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